EEE104 – Digital Electronics (I)

Lecture 8

Dr. Yuxiao Fang

Emerging Electronics and Energy Technologies Research Centre (EEET)

Office: SC242

Office hour: 1:00-5:00 pm, Wednesday

In This Session

- Combinational Logic
  - Basic Combinational Logic Circuits
  - Implementation
  - Universal NAND and NOR Gates
  - Using NAND and NOR Gates

Basic Combinational Logic Circuits

- A **combinational logic** circuit is one in which
  - Logic gates are connected together to implement a certain function (in contrast to *basic logic gates*).
  - There is no storage involved (in contrast to *sequential circuits*).
- Examples of combinational logic
  - Exclusive-OR gates
  - The AND-OR logic in an SOP implementation.

The settled output depends on the current inputs.

Basic Combinational Logic Circuits

Exclusive-OR Logic (XOR)

The output is **HIGH** when the two inputs are at **opposite** levels.

$$X = A\overline{B} + \overline{A}B = \mathbf{A \oplus B}$$

A	B	XOR	XNOR
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

Exclusive-NOR Logic (XNOR)

Invert the XOR output: **HIGH** when the two inputs are at the **same** level.

$$X = \overline{A\overline{B} + \overline{A}B} = (\overline{A} + B)(A + \overline{B})$$

$$= \overline{A}\,\overline{B} + AB$$

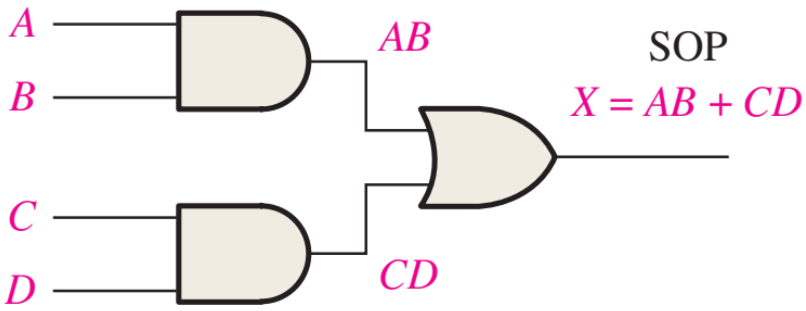
$$= \overline{\mathbf{A \oplus B}}$$

Basic Combinational Logic Circuits

AND-OR Logic

The outputs of AND gates connect to an OR gate.
This implements an **Sum of Products (SOP)** expression.

$$X = AB + CD$$



AND-OR-Invert implements a **POS (Products of Sum)** expression:

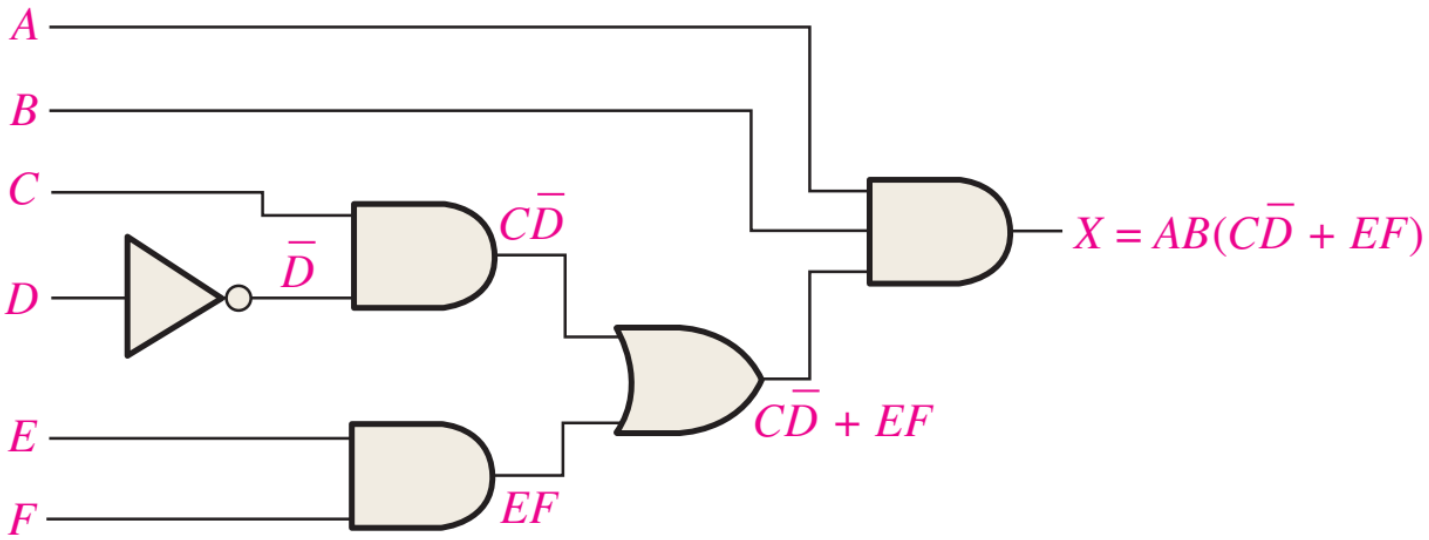
$$\overline{AB + CD} = \overline{AB}\,\overline{CD}$$

$$= (\overline{A} + \overline{B})(\overline{C} + \overline{D})$$

Implementation

From a Boolean Expression to a Logic Circuit

$$\mathbf{X = AB(C\overline{D} + EF)}$$



$$X = AB(P + Q) \quad P = C\overline{D} \quad Q = EF$$

Practice

A = B = C = D = F = 1 and E = 0. Find X.

Universal NAND and NOR Gates

- The NAND and NOR gates are **universal** because they can be used to produce any other logic function

NAND gate	**NOR gate**
– NOT	– NOT
– AND	– AND
– OR	– OR
– NOR	– NAND

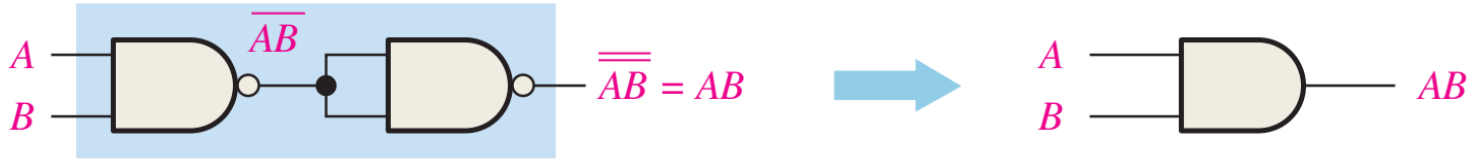
NOT, AND and OR are sufficient to implement any Boolean function.

Universal NAND Gates

NAND Gate as an Inverter



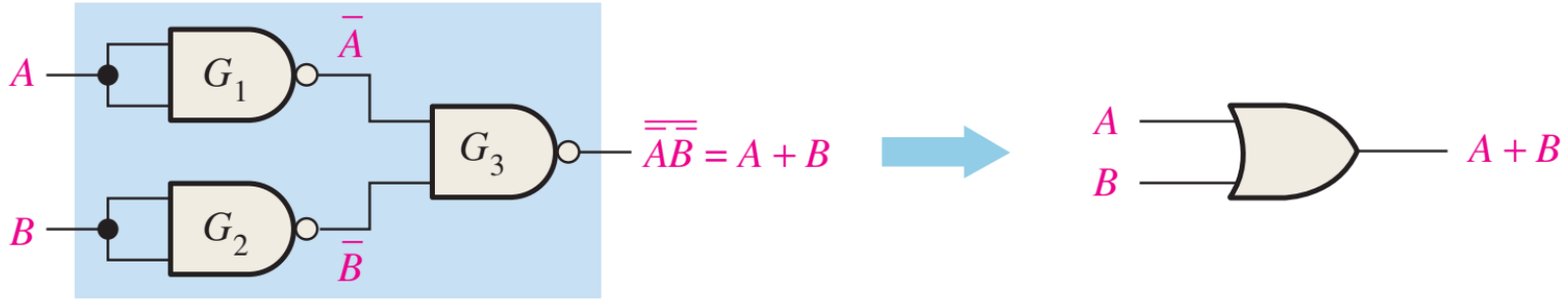
Two NAND Gates as an AND Gate



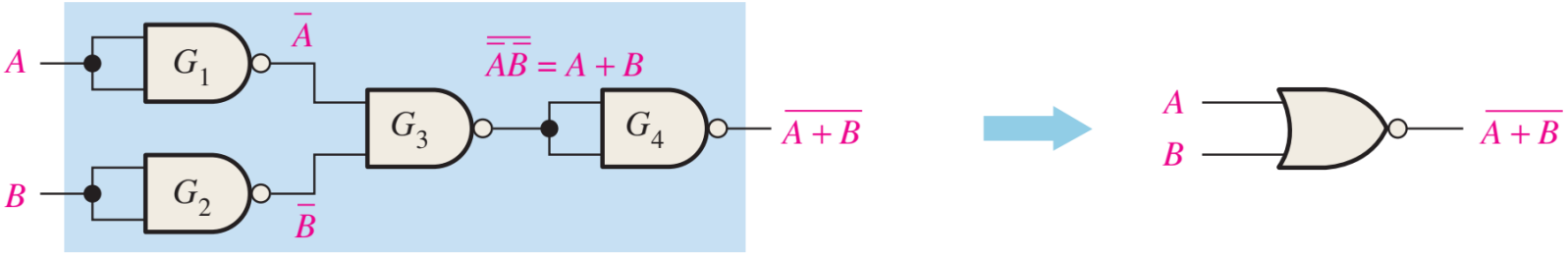
The second NAND removes the first output inversion.

Universal NAND Gates

Three NAND Gates as an OR Gate



Four NAND Gates as a NOR Gate



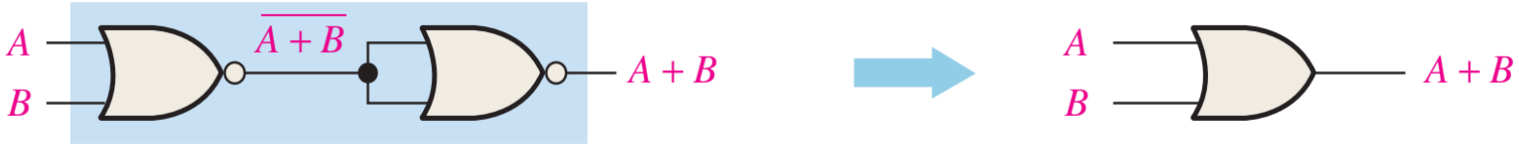
One extra inversion changes OR into NOR.

Universal NOR Gates

NOR Gate as an Inverter



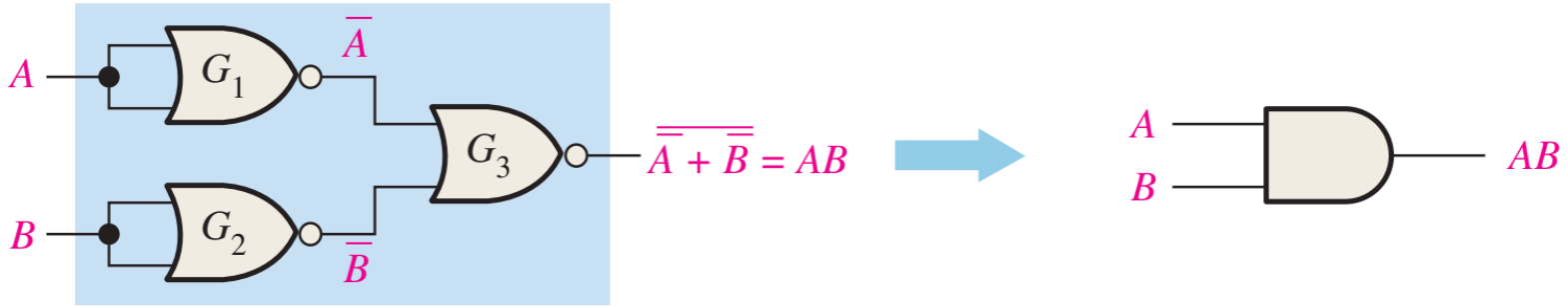
Two NOR Gates as an OR Gate



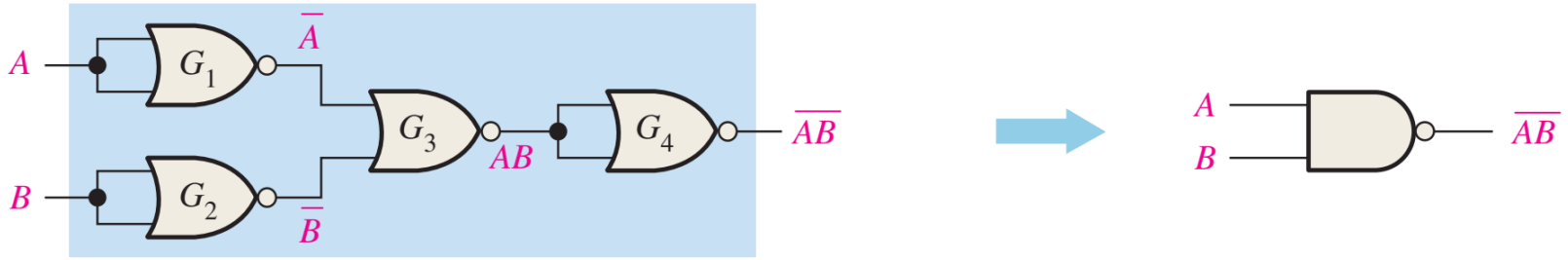
The second NOR removes the first output inversion.

Universal NOR Gates

Three NOR Gates as an AND Gate



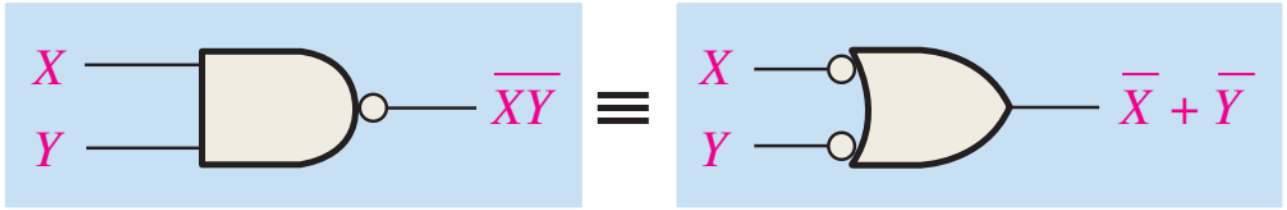
Four NOR Gates as a NAND Gate



One extra inversion changes AND into NAND.

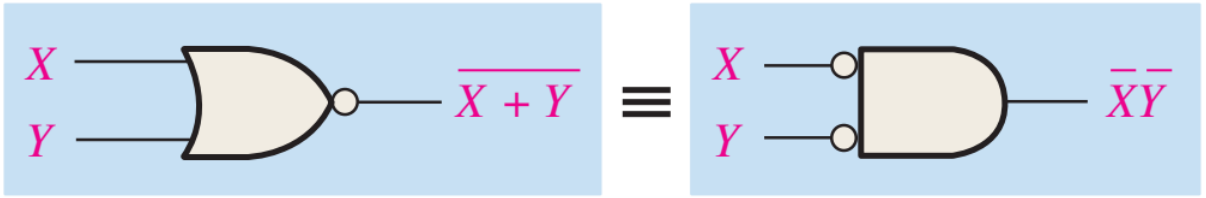
Equivalent NAND and NOR Gates

A NAND gate is equivalent to a negative-OR gate.



NAND Negative-OR

A NOR gate is equivalent to a negative-AND gate.

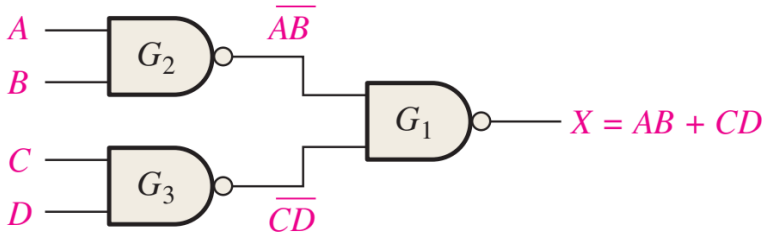


NOR Negative-AND

Appropriate equivalent symbols make a logic diagram easier to read.

Equivalent NAND and NOR Gates

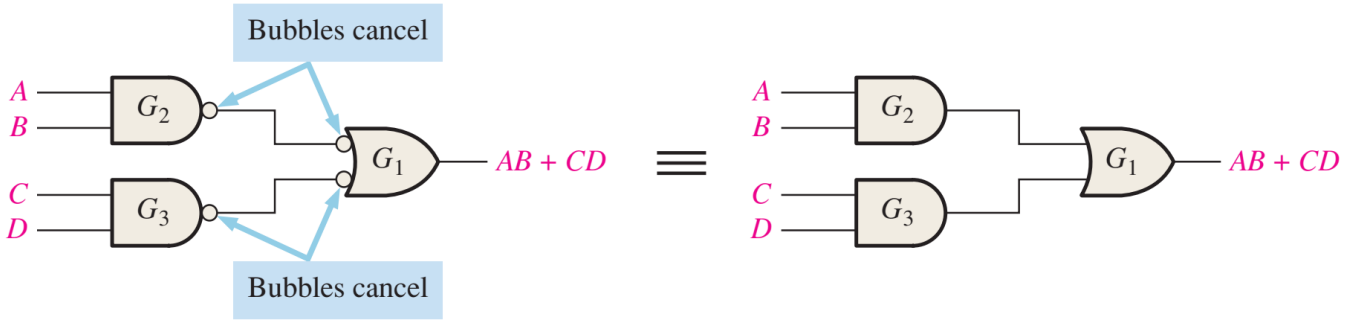
Read the output of each gate, then apply DeMorgan's theorem.



$$X = \overline{\overline{AB}\,\overline{CD}}$$

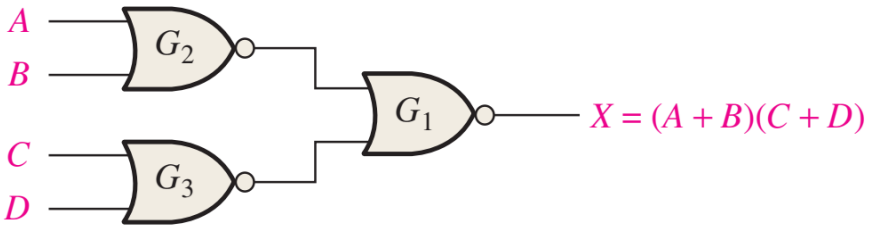
$$= \mathbf{AB + CD}$$

Use the negative-OR symbol for G1: the paired bubbles cancel.



Equivalent NAND and NOR Gates

Read the output of each gate, then apply DeMorgan's theorem.



$$X = \overline{\overline{A + B} + \overline{C + D}}$$

$$= (A + B)(C + D)$$

Use the negative-AND symbol for G1: the paired bubbles cancel.

